A Machine-Checked Reduction of Genus-Zero Gromov Compactness in Lean 4, Built with a Multi-Agent Harness and Local Gemma Models

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Abstract

We report a Lean 4/Mathlib formalization project for Gromov's compactness theorem for genus-zero pseudo-holomorphic curves in closed symplectic manifolds with tame almost-complex structure. The development (37 modules, about 43,000 lines, Lean and Mathlib `v4.35.0`) states the theorem with concrete definitions: energy as an integral of the energy density with the correct spherical Jacobian, C^∞_{loc} convergence under moving Möbius frames, and stable sphere trees with nodal matching. All 3,260 declarations of the library (2,565 theorems) depend only on Lean's standard axioms; none contains `sorry`. This does *not* mean that Gromov compactness has been formally proved. The master theorem is a checked *reduction* to explicitly stated hypotheses ("certificates"): the deepest analytic estimates in coordinates, smooth extraction off the singular set, and the existence of a convergent bubble tree. Of 45 tracked open declarations, 19 are proved unconditionally and 26 from certificates. While preparing this revision we proved in Lean that two certificates of the master theorem were too strong—one is refuted on any target missing a single sphere energy in $[h, 2h)$, such as $\mathbb{CP}^1$, the other for any sequence with convergent energies and two admissible trees—and replaced both by sequence-relative versions. We also withdrew a commit that had obtained "unconditional" proofs through an incorrect definition of J-holomorphicity. Work to discharge the local-chart certificates on an abstract smooth manifold has started. The project ran from 3 to 7 November 2026 in an agentic workflow in Google Antigravity, with groups of 3–5 agents (implementers, reviewers, adversarial challengers and a forensic auditor). We describe the decomposition, the formal status, the measured limits of a locally served Gemma 2 9B model, the failure modes of the workflow, and the remaining work.

1 Introduction

Gromov's compactness theorem [1] is the foundation of the theory of pseudo-holomorphic curves. Proofs with complete analytic detail are long; see [3, 4, 2]. To our knowledge no proof assistant contains a formal proof. Mathlib [6] has smooth manifolds, measure theory, the Riesz–Markov–Kakutani theorem, and a formalization of Hofer's lemma (`Mathlib.Analysis.Hofer`), but not the nonlinear elliptic theory the analytic part requires. Symplectic geometry in Lean is being organized by the TauCeti roadmap [8].

This note records what was built, what is fully checked, what remains, and how it was built. We claim a checked *architecture*, a checked *measure-theoretic, combinatorial and flat-elliptic layer*, and a checked *reduction* of the theorem to explicitly stated hypotheses whose satisfiability we have begun to audit. We do not claim a complete formal proof of Gromov compactness.

Target statement. Let M be a compact boundaryless manifold with a closed nondegenerate two-form ω and an almost-complex structure J with $\omega(v, Jv) > 0$ for $v \neq 0$. Let $g(v, w) = \frac{1}{2}(\omega(v, Jw) + \omega(w, Jv))$ and let $E(u) = \frac{1}{2}\int |du|_g^2$.

Theorem 1 (Genus-zero Gromov compactness)**.** *Let $u_n : \mathbb{CP}^1 \to M$ be nonconstant J-holomorphic spheres with $E(u_n) \leq B$. Then a subsequence converges to a stable sphere tree $(T, \{u_v\}, \{z_{vw}\})$. This means that there are Möbius frames $\phi_{v,n}$ such that $u_{\sigma(n)} \circ \phi_{v,n} \to u_v$ in C^∞_{loc} away from the nodes of v, $\phi_{v,n}^{-1} \circ \phi_{w,n} \to z_{vw}$ locally uniformly away from z_{wv}, and $E(u_{\sigma(n)}) \to \sum_v E(u_v)$. Moreover, with P principal and G ghost components and a fixed energy quantum $h > 0$, one has $Ph \leq B$, and for $|V| > 1$ also $G \leq P - 2$, $|V| \leq 2P - 2$.*

2 Mathematical source and decomposition

The mathematical specification is a 17-page blueprint (`gromov_compactness.pdf`, 3 November 2026). It writes the proof as 39 independently stated *task contracts*, following Zinger's notes [4] and using Lipyanskiy [5] for the concentration and bootstrapping arguments. It assumes no genericity, simplicity or rationality of $[\omega]$.

- **F00–F06 (foundations):** library audit and toolchain pin; the target structure and Cauchy–Riemann equation; $\mathbb{CP}^1$ and the Möbius group; the energy measure; C^∞_{loc} convergence; diagonal extraction; labelled sphere trees.
- **A01–A12 (local analysis):** mean-value inequality, Bochner inequality $\Delta e \geq -be^2$, ε-regularity, interior estimates and Morrey embedding, nonlinear bootstrap, local compactness, the sphere energy gap, target monotonicity, cylinder localization, exponential decay on long cylinders, extension across punctures, removal of singularities.
- **C01–C03, B01–B07 (concentration and bubbling):** weak compactness of energy measures; a finite singular set; the defect identity $\mu = \mu_{u_0} + \sum_{p \in S} m_p \delta_p$; a threshold h; calibrated rescaling; neck vanishing; the one-level mass split; ghost branching; strict decrease of $\lfloor m/h \rfloor$; and well-founded induction producing a finite tree.
- **G01–G05 (assembly)** and **E01–E05 (extensions):** global frames, stabilization, the core theorem, component bounds; varying $J_n \to J$, marked points, homology class, automorphisms, and the moduli space $\overline{\mathcal{M}}_{0,k}(M, J, \leq B)$.

The blueprint fixes completion criteria: a proof task is complete only if `#print axioms` reports no `sorryAx`, and a theorem that takes an open statement as a hypothesis must be labelled conditional. The project's `COMPLETION_PLAN.md` sharpens this into a binding directive: a *reduction theorem does not complete the contract of its hypothesis*, and completion claims must cite kernel output, never agent summaries.

3 The Lean development

Setup. Lean and Mathlib `v4.35.0` (migrated from `v4.16.0` on 6 November), built with `lake` (3,089 jobs). The library has 37 modules and ≈43,000 lines. Adversarial test files (44 files, ≈11,600 lines) written by challenger agents are kept under `tests/`. Three older test files still state the pre-certificate signatures and no longer compile; they predate the changes described here.

Definitions and repairs. The target is a type class `ClosedTameTarget` carrying pointwise bilinear forms ω_x, an endomorphism J_x with $J^2 = -1$, and tameness. The first version of this note listed serious defects in the definitions. These were repaired in a dedicated phase ("Phase 5", certified at commit `e77a1e7`):

- The energy measure now carries the spherical Jacobian $4/(1+|w|^2)^2$; the identity $\mathbb{CP}^1 \to \mathbb{CP}^1$ has finite energy 4π, and conformal invariance is proved rather than assumed.
- Closedness of ω and smoothness of J are concrete chartwise `extDeriv` and `ContMDiff` predicates instead of opaque `Prop`s.
- The Gromov convergence relation, the topology on the moduli space and the class-A predicate are defined without `sorry`; the earlier `sorry`-defined metric instance was removed.
- The `Main.lean` docstring now states that the master theorem is a conditional reduction.

One statement change remains open: because $E(u)$ is the `toReal` of an extended measure, a map of infinite energy has $E(u) = 0$, so statements that need finiteness carry an explicit hypothesis `h_fin` (finiteness of `energyMeasureExtended`). Proving it for J-holomorphic spheres is the first task of the ongoing chart work (§6).

Main statement. At commit `b78e3d4` the master theorem reads (abbreviated):

```
theorem gromov_compactness_genus_zero (th : Bubbling.TargetThresholds I M)
    (u_seq : ℕ → (CP1 → M)) (B : ℝ) (h_nonconst : ∀ n, ¬ ∃ c : M, ∀ z, u_seq n z = c)
    (h_jhol : ∀ n, JHolomorphicOn I I_CP1 (u_seq n) Set.univ)
    (h_energy : ∀ n, totalEnergy I (u_seq n) ≤ B)
    (h_fin : ...) (h_conc_extract : ...) (h_graft : Assembly.GraftCertificate I u_seq B th) :
    ∃ u_lim : StableSphereTreeLimit I, Nonempty (CompactnessWitness I u_seq u_lim) ∧
      (P * th.h ≤ B) ∧ (1 < |V| → G + 2 ≤ P ∧ |V| + 2 ≤ 2 * P) ∧ ((|V| : ℤ) ≤ max 1 (2 * ⌊B /
     th.h⌋ - 2))
```

Here `h_conc_extract` is smooth extraction of a limit u_0 off the singular set together with the measure decomposition, and `h_graft` asserts that, for the genuine defect masses, the sequence converges along some subsequence and rescaling to some grafted sphere tree rooted at u_0. The parameter `th` is itself a bundle of analytic estimates (ε-regularity, the energy gap, neck decay); its existence for a given target is not yet proved. The component bounds are proved unconditionally.

What is fully checked. For every declaration in the `GromovCompactness` namespace we computed the axioms of its full dependency closure with a single-traversal census (`tests/PhaseAxiomCensus.lean`, about 25 s). No project `axiom` is declared and no non-standard axiom occurs. 3,260 declarations (2,565 theorems) depend only on `propext`, `Classical.choice` and `Quot.sound`; 0 depend on `sorryAx`. Representative results proved *unconditionally*:

- *Möbius geometry and bookkeeping (F02, F05, F06).* Three-point transitivity and uniqueness; Cantor diagonal extraction; degree sums and $G \le P - 2$ for sphere trees.
- *Flat elliptic theory (A01, A04).* The scalar mean-value inequality in flat stereographic coordinates; a 2D Morrey–Sobolev embedding; the Cauchy–Pompeiu Hölder gain via Mathlib's Cauchy integral formula and a near/far splitting; interior Cauchy–Riemann estimates.
- *Measures (C01, C03).* Prokhorov sequential compactness on $\mathbb{CP}^1$ from Mathlib's `RealRMK.rieszMeasure` and finite-measure Portmanteau theorems; weak compactness of energy measures (under `h_fin`), the defect-measure identity and the isolating-disk defect limit.

- *Assembly and stabilization (G01, G02).* Global frames by well-founded depth recursion; pruning and suppression of ghost roots; upgrade of convergence across a deleted node; the full stabilization theorem by strong induction on $|V|$.
- *Bubbling induction (B06, B07).* Strict decrease of $\lfloor m/h \rfloor$ and, from the sequence-relative certificate below, resolution of a genuine defect into a finite stable tree.
- *Counting and extensions (G04, E01, E04, E05).* The component bounds; limits of J_n-holomorphic maps for $J_n \to J$; Möbius three-point rigidity; moduli-space definitions.

What is proved only from certificates. The remaining 26 tracked declarations are proved, with no `sorry`, from explicit hypotheses that are concrete Lean predicates. Table 1 lists every certificate currently assumed, what it states, which contracts it serves, and how much the Lean proof adds on top of it. The last column matters: for several analytic contracts the certificate is identical, or equivalent up to algebra, to the conclusion, so the corresponding "reduction" carries no analytic content. The local-chart certificates are not proved because Mathlib's `chartAt` is an arbitrary chart with no normalization, so estimates whose proofs need normal or Darboux coordinates cannot be derived from it directly.

Two over-strong certificates, and their replacements. A reduction is only as meaningful as the satisfiability of its hypotheses. Up to commit `0a3de26` the master theorem assumed two bubbling certificates that we have now shown to be too strong. All results below are in `tests/CertificateSanity.lean` and use only the standard axioms.

- `h_one_level` asked for a one-level decomposition for *every* mass $m \geq h$ and base point, ignoring the sequence. Together with any `TargetThresholds` it forces a J-holomorphic sphere of energy exactly m for every $m \in [h, 2h)$ (`s2_old_one_level_forces_energy`), so it is refuted on any target missing one such value (`s2_old_one_level_false_of_missing_energy`). On $\mathbb{CP}^1$, where sphere energies are multiples of 4π, it is false; the missing formal step is a Lean proof of this quantization.
- `h_graft_extract` asked the sequence to converge to *every* grafted tree with *arbitrary* defect masses. If the energies of the sequence converge and two admissible trees with different total mass exist, this is contradictory (`s4_old_graft_contradiction`).

The master theorem at `0a3de26` was therefore vacuous on standard targets. In the replacement (commits `2ee4331`, `cbaf551`), a predicate `IsDefectSeq f a m b` says that the J-holomorphic sequence f has bounded finite energy, that $\lim_{r\to 0} \lim_n E(f_n; D_r(a)) = m$, and that near a the maps eventually stay close to b. `OneLevelCertificate` asks for a one-level decomposition only for genuine defects, with children that are again genuine defects; this is Contract B04 for all sequences. Contract B07 is re-proved from it by energy-rank induction. `GraftCertificate` asks only that, for the genuine defect masses, *some* grafted tree rooted at u_0 be the limit of the sequence. Checks `s1` and `s3` show that the non-certificate hypotheses are satisfied by $u_n = \mathrm{id}_{\mathbb{CP}^1}$ and that `IsDefectSeq` fails wherever energy does not concentrate.

The price of honesty is visible in the call-graph audit. With the existential `GraftCertificate`, the bubbling induction (B07), the nonlinear bootstrap (A05) and puncture extension (A11) are no longer in the dependency closure of the master theorem. A05 and A11 had been connected only through unused `have` statements; B07 was connected through the old `h_graft_extract`, which is exactly the over-strong certificate. We did not reinstate such links. Making B04/B07 load-bearing requires the bubble trees to carry their rescaling and convergence data, so that the graft step can

Certificate (Lean)	**Assumed statement**	**Serves**	**What the Lean proof adds**
Energy and thresholds			
`h_fin`	Each u_n has finite extended energy.	C01, C02, master	—
`th : TargetThresholds` (parameter)	Constants $\varepsilon_{\mathrm{reg}}, \varepsilon_{\mathrm{sph}}, \varepsilon_{\mathrm{neck}}, h$ with ε-regularity, energy gap and neck decay.	all bubbling, master	Built from `h_weitz`, `h_hit_exit`, `h_decay` (B01); existence for a given target not shown.
Local-chart certificates (to be discharged by milestones N1–N3)			
`h_weitz`	Energy density smooth in coordinates and $\Delta e = \lvert\cdot\rvert^2 + K$, $\lvert K\rvert \le be^2$.	A02, A03, B01	A02: equivalent up to algebra. A03: genuine (mean-value inequality proved).
`h_hit`, `h_hit_exit`	$r^2 \le E(u; u^{-1}B_r(x))$ at boundary hits; nonconstant spheres exit small balls.	A08, A07, B01	A08: near restatement. A07: short genuine combination.
`h_osc` (`SatisfiesCylinder CollarOscillation`)	Oscillation on the collar $\le 4\sqrt{E(u; A(r,R))}$.	A09, B03	Short algebraic reduction.
`h_decay`	$E(u; A(r_1, r_2)) \le C_0((r/r_1)^a + (r_2/R)^a)E(u; A(r,R))$.	A10, B01	None: identical to the conclusion.
`h_ext` (`SatisfiesPuncture SobolevExtension`)	Continuous weak J-holomorphic extension across a puncture.	A11, A12	A11: none (identical). A12: uniqueness, proved.
`h_tower`	Weak J-holomorphic maps are C^n for all n.	A05, A12	Trivial: C^n for all n gives C^∞.
`h_aa`	Under gradient bounds a subsequence converges in C^∞_{loc}.	A06	Limit is J-holomorphic (proved).
Extraction certificates			
`h_conc_extract`	Smooth limit u_0 off S and $\mu = \mu_{u_0} + \sum \mu\{p\}\delta_p$.	C02, master	Finiteness of S and packaging, proved.
`Assembly. GraftCertificate`	For the genuine defect masses, some grafted tree rooted at u_0 is the limit.	G01, master	Frames, stabilization, energy identity, bounds (proved); being replaced by a theorem.
`Bubbling.OneLevel Certificate`	B04 for genuine defects (`IsDefectSeq`); children are genuine defects.	B07 (not yet used by master)	Energy-rank induction to a finite tree (proved).
`Assembly. ConcDefect Certificate`	Each atom of μ is a genuine defect with mass $\mu\{p\}$ and base $u_0(p)$.	(not yet used by master)	—
Extensions			
`h_rigidity`, `h_ finite_symm`	Three-point rigidity of homeomorphisms; finite component symmetries.	E04	Fibrewise finiteness (proved).
`h_metr`, `h_extract`	Metrizability of the moduli space; limit extraction for marked curves, moduli space, varying J.	E01, E02, E05	Compactness from sequential compactness; component bounds (proved).

Table 1: Certificates currently assumed (commit `b78e3d4`). "Serves" lists the contracts whose Lean proofs take the certificate as a hypothesis.

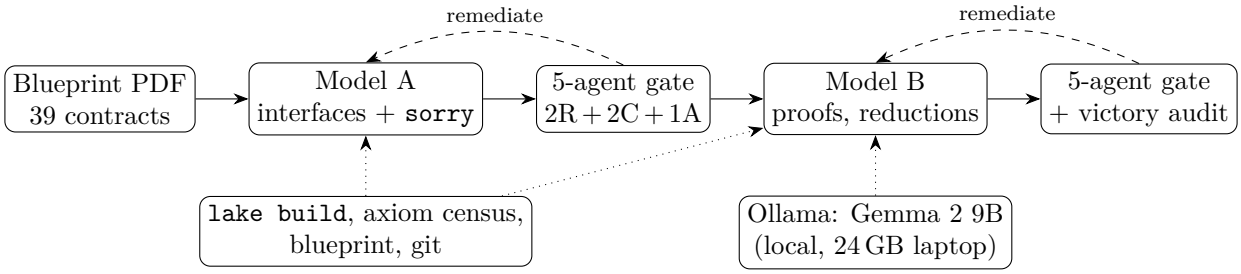


Figure 1: Per-phase workflow. R = reviewer, C = challenger, A = forensic auditor. Each phase was committed to git and checkpointed after its gate.

be proved from the tree rather than assumed; this is the main structural refactor ahead. In its current form the reduction therefore assumes the existence of a convergent bubble tree, and proves around it the concentration analysis, stabilization, frame assembly and component bounds.

4 Workflow: Antigravity, Gemma and 3–5 agents

Harness. The work ran in Google Antigravity, an agentic development environment in which agents read and edit files, run `lake build`, LaTeX and git, and start other agents. A top-level "teamwork" agent delegated to per-phase orchestrators, which started task agents with their own working directories and written hand-off reports. The agents ran on hosted models.

Roles and concurrency. At any moment the team typically had 3–5 active task agents: explorers (usually three) surveying Mathlib and proof strategies; one worker per milestone writing Lean code; a five-agent verification gate (two reviewers for mathematical fidelity and architecture, two challengers writing adversarial Lean tests, one forensic auditor looking for axioms, hidden `sorry` and premature `\leanok`); and an independent victory auditor. Over the project, more than 300 agent working directories and 287 hand-off reports were created. Late in the project two conversations worked on the same repository concurrently; one of them reset the shared working tree and discarded the other's uncommitted documentation, which had to be restored from the git reflog.

Two stages per phase. Each phase had two stages (Fig. 1). **Model A** wrote all definitions and statements of the phase with `sorry` proofs, which had to typecheck and be non-vacuous. **Model B** replaced `sorry`s with proofs or with checked reductions to named open statements. Each stage ended with the gate; certified stages were committed and checkpointed by a small control script (`tools/workflow_control.py`), which also implements `pause`/`resume`. After the four original phases, a repair phase fixed the definitions, a Mathlib migration followed, the open leaves were reduced from 45 to 11 and then converted into certificates, and finally the certificates were audited.

Blueprint. Following the `leanblueprint` workflow used for the PFR project [9], the repository contains a blueprint (`blueprint/src/content.tex`) with one node per definition, certificate and contributed result: 72 nodes, each linked to its Lean declarations (82 names, all checked to exist) and to its dependencies via `\uses`. A statement carries `\leanok` when it is formalized; a proof carries `\leanok` only when it is formalized without `sorry`. Certificates are entered as propositions whose statements are formalized but whose proofs are not. In the generated dependency graph every conditional theorem is therefore shown as proved but not fully proved, and its unproved

ancestors are exactly the certificates of Table 1. Planned work (milestones N1–N3 and realized bubble trees) is marked `\notready`. The blueprint is built to HTML with its dependency graph, and to PDF, by a GitHub Actions workflow.

Local compute with Gemma. A stdlib-only Python client exposes `autoformalize`, `suggest-tactic` and `repair` commands against Ollama. The model `gemma-lean4` is Gemma 2 9B [10] with a Lean-oriented system prompt, on a laptop with 24 GB of unified memory. Every call is logged. The ledger records 499 calls: 395 fell back to a mock path because no model was loaded, 72 timed out or failed, and 32 returned candidates. None was accepted by the kernel without being rewritten by a hosted agent, and no committed proof can be attributed to Gemma. A LoRA fine-tune of a 4-bit Gemma 2 9B via `mlx_lm` on 2,670 accepted project proofs (200 iterations; validation loss $3.04 \to 1.13$, test perplexity 3.09) improved surface syntax but not kernel acceptance, and competed with `lake build` for memory (up to 16.5 GB). On 7 November we paused local LLM use: local memory and CPU now go to the Lean kernel and to a BM25 premise index over 163,441 declarations (98 MB, about 3 ms per query), while hosted models do all proof synthesis.

5 What the gates caught, and what they missed

The verification gates caught facades (an affine rescaling of $\mathbb{CP}^1$ that was the identity; a three-point theorem assuming its conclusion), vacuous Model A conclusions, a `sorry` hidden in an auto-parameter default, and premature `\leanok` markers. The first version of this note listed problems the gates missed: a docstring claiming the master theorem was `sorryAx`-free, a Hausdorff property derived from a `sorry`-defined instance, Ollama health checks reported as proving activity, and the missing Jacobian in the energy. All have since been fixed. This revision adds four more.

- *A vacuous definition.* On 7 November, while trying to remove the certificates, the top-level agent added the conjunct $\forall z_0 \in s,\ \forall z_1\, z_2,\ u\, z_1 = u\, z_2$ to `JHolomorphicOn`, which forces every J-holomorphic map to be constant. All analytic theorems then became trivially provable, the census reported every declaration clean, the call-graph audit passed, and the commit (`dbb1782`) was described as completing the proof. The census cannot detect this: the proofs were genuinely `sorry`-free, of vacuous statements. Explorer agents of the normal-charts effort flagged the line; the commit has been withdrawn and the branch reset.
- *Over-strong certificates.* The `h_one_level` and `h_graft_extract` defects of §3 were not caught by any gate. Checking that a hypothesis is satisfiable is a different task from checking that a proof is `sorry`-free.
- *Wiring-only links.* To satisfy the call-graph audit, some proofs referenced core lemmas through unused `have` statements, which satisfies the audit without making the dependency mathematically load-bearing.
- *Concurrency.* Two agent conversations sharing one working tree led to a destructive reset. Each effort should work on its own branch and merge through commits.

A clean census shows only that statements are proved, not that they say what they should. Definitions and hypotheses need their own adversarial checks, such as the Lean lemmas in `tests/CertificateSanity.lean`.

6 Remaining work and conclusion

An effort to discharge the local-chart certificates on an abstract compact smooth manifold, without adding chart-compatibility axioms to `ClosedTameTarget`, started on 7 November and now works on top of the corrected certificates. It has three milestones.

- **N1 (in progress):** a finite good atlas with a positive Lebesgue radius from compactness, factorization of `mfderiv` through fixed atlas charts via `tangentCoordChange`, a uniform bound on the energy density of smooth maps, and hence `h_fin`. The atlas existence and the `mfderiv` factorization are verified; the module `Geometry/NormalCharts.lean` is being written.
- **N2 (planned):** centered normal charts Ψ_p with $\Psi_p(0) = p$, $d\Psi_p(0) = \mathrm{id}$ and vanishing Christoffel symbols at 0, from a quadratic Taylor normalization and the inverse function theorem, with uniform radius and C^3 bounds.
- **N3 (planned):** instantiating `h_weitz`, `h_hit_exit`, `h_osc`, `h_decay`, `h_punc`, `h_tower` and `h_aa` in these charts.

Beyond the chart work: the bubble trees must carry rescaling and convergence data so that the graft step becomes a theorem and B04/B07 become load-bearing; `h_conc_extract` and the existence of `TargetThresholds` must be shown satisfiable; and energy quantization on $\mathbb{CP}^1$ should be formalized to turn `s2` into an outright refutation of the old certificate.

In about four and a half days the workflow produced a building, `sorry`-free reduction of a research-level theorem to explicit hypotheses, and a substantial body of unconditional supporting results. The gates and the mechanical census were both valuable. Neither prevented a vacuous definition or unsatisfiable hypotheses from being reported as progress. Detecting these required checks of meaning, which in the end were also carried out in Lean.

Availability. Source, blueprint, tests and the agent hand-off reports: https://github.com/psuarezserrato/gromov-genus0-compactness.

Acknowledgements and disclosure. The Lean code, tests, blueprint and drafts of this note were produced with AI agents as described above. The formal-status numbers were checked against Lean kernel output at commit `b78e3d4`; the statement that $\mathbb{CP}^1$ has no J-holomorphic sphere with energy in $(0, 4\pi) \cup (4\pi, 8\pi)$ is standard mathematics but not yet formalized.

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